Figure 3 round 2 · decomposition and censoring

At matched edge gap, larger native size accounts for most of the increase in driven handover time. Extra accessible separation contributes relatively little. Cue-free competition has a stronger separation response and remains unresolved in several mass-conserved-model conditions.

For each model and assay, the matched gap is the largest gap within both measured size curves. In this dataset it is the small-size antipodal-placement endpoint. In B, the coordinate is the measured gap at withdrawal, not the initial cue gap. The endpoint called antipode is the trajectory seeded by the initial antipodal cue: its withdrawal region centers can shift.

A1 uses native continuous complete-region handover roots. B uses saved-sample lower and upper time brackets. Linear interpolation is restricted to adjacent measured eligible placements. Interval endpoints are interpolated separately; any censored upper endpoint stays infinite. Missing or topology-ineligible values are never converted into bounds.

A ratio with a censored numerator and resolved denominator has a lower bound. A resolved numerator and censored denominator gives an upper bound. Two censored times give no finite one-sided ratio bound; the result is marked indeterminate. In particular, dividing two horizon caps is invalid. Finite bars use the geometric center of the resulting interval, with the full interval recorded in CSV.

The exact product identity uses the same interpolated matched-gap time in both terms. For finite B intervals the representative values obey the identity; multiplying independently propagated marginal interval limits would ignore this shared denominator and numerator. Total intervals are computed directly from endpoint times.

Optional extensions awaiting user review: 9 Holmes ratio-1 histories (5 small, 4 middle), each completed to 20,000 and eligible for a proposed 80,000 horizon. They lie within the common withdrawal-gap range. No extension has been submitted. These extensions may still leave large-size matched-gap or antipodal times censored.

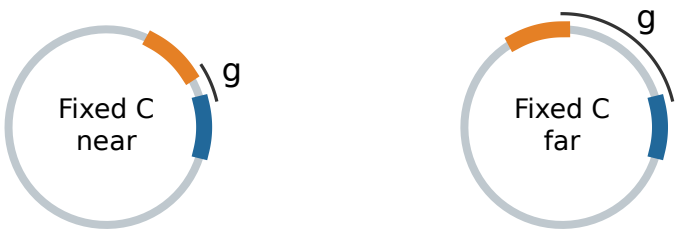
The following pages retain BOTH superseded main pages, including the complete translation page, then the entire seven-page existing supplement. All four native-profile appendix volumes remain linked in the existing reader; no model inputs, histories, or prior supplements are replaced.

All decomposition values · time ratios with interval bounds

Model	Assay	Size / small	Matched gap	Size term	Extra separation	Total
Wave-pinning	A1	37.8 / 30	7.719	2.14	1.01	2.17
Wave-pinning	A1	47.6 / 30	7.719	5.59	1.05	5.86
Goryachev	A1	37.8 / 30	12.05	1.84	1.04	1.91
Goryachev	A1	47.6 / 30	12.05	3.5	1.15	4.02
Holmes	A1	37.8 / 30	6.29	3.59	1	3.6
Holmes	A1	47.6 / 30	6.29	23.9	1.01	24.2
Spring	A1	3.15 / 2.5	0.4296	2.44	1	2.44
Spring	A1	3.97 / 2.5	0.4296	8.45	1.01	8.52
Wave-pinning	B	37.8 / 30	8.378	5.67–5.77	1.02–1.03	5.81–5.92
Wave-pinning	B	47.6 / 30	8.378	≥15.9	indeterminate (both censored)	≥15.9
Goryachev	B	37.8 / 30	11.87	1.15–1.16	1.01–1.02	1.16–1.18
Goryachev	B	47.6 / 30	11.87	1.44–1.47	1.06–1.09	1.54–1.58
Holmes	B	37.8 / 30	7.609	indeterminate (both ce	indeterminate (both censored)	indeterminate (both censored)
Holmes	B	47.6 / 30	7.609	indeterminate (both ce	indeterminate (both censored)	indeterminate (both censored)
Spring	B	3.15 / 2.5	0.5456	4.2–4.38	1.73–1.77	7.31–7.71
Spring	B	3.97 / 2.5	0.5456	≥6.87	indeterminate (both censored)	≥13.8

Intervals reflect saved-time brackets, not biological uncertainty. ≥ is a one-sided lower bound.
All interval endpoints, interpolation weights and source slots appear in the companion CSVs.

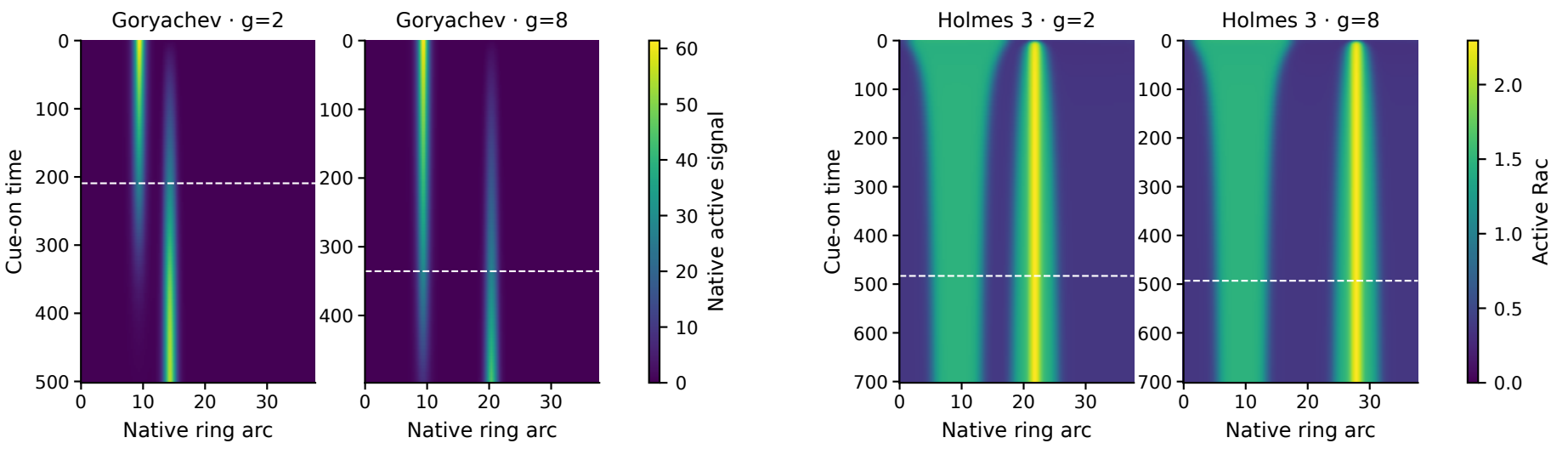
Figure 3 · Separation affects native competition differently across models



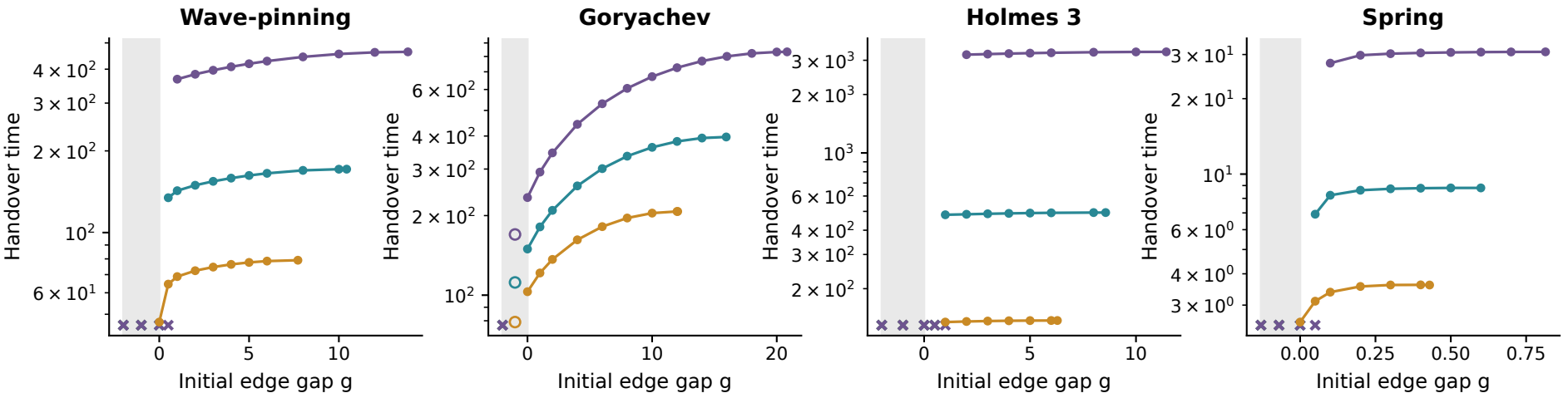
A Rotate one native cue at fixed size

g: initial incumbent 10% edge → cue near edge
A1: continuous cue → complete-region handover
A2: own-handover-scaled pulse → cue-free fate
B: seed ratio 0.5 / 1 / 2 → cue-free peak resolution
12 fresh qualified basals; native kinetics unchanged

B Same-size native histories · white line: complete-region handover



C1 Driven handover · open symbols: borderline placements



C2 Cue-free resolution · ratio 1 · triangles retain finite-cap lower bounds

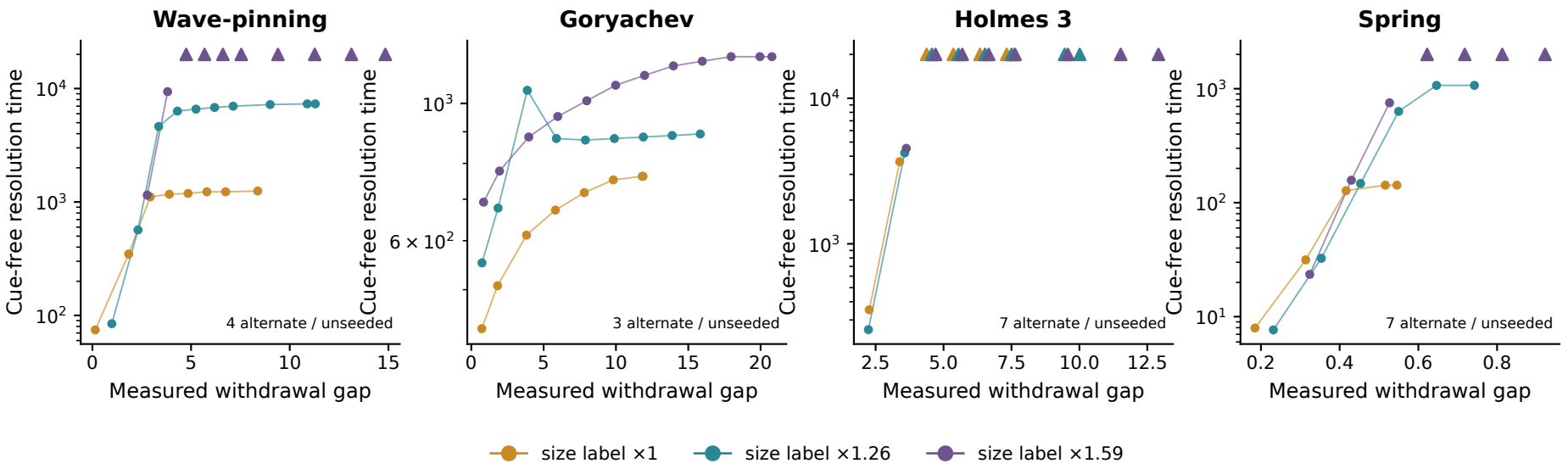


Figure 3 · Geometry remaps challenges; activity direction is measured separately

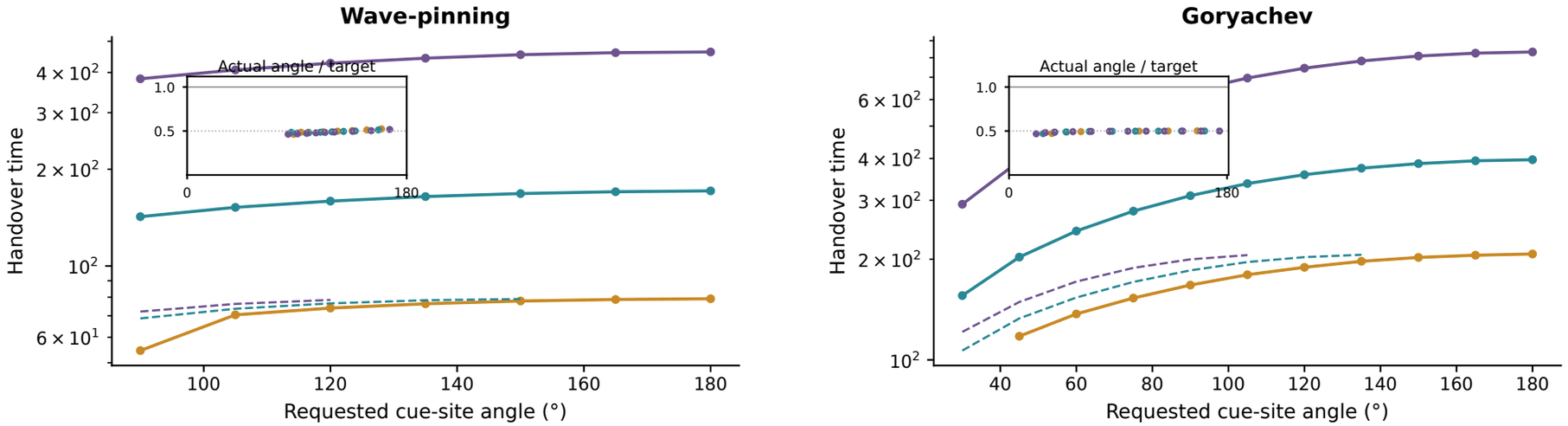


D Physical gap and center angle

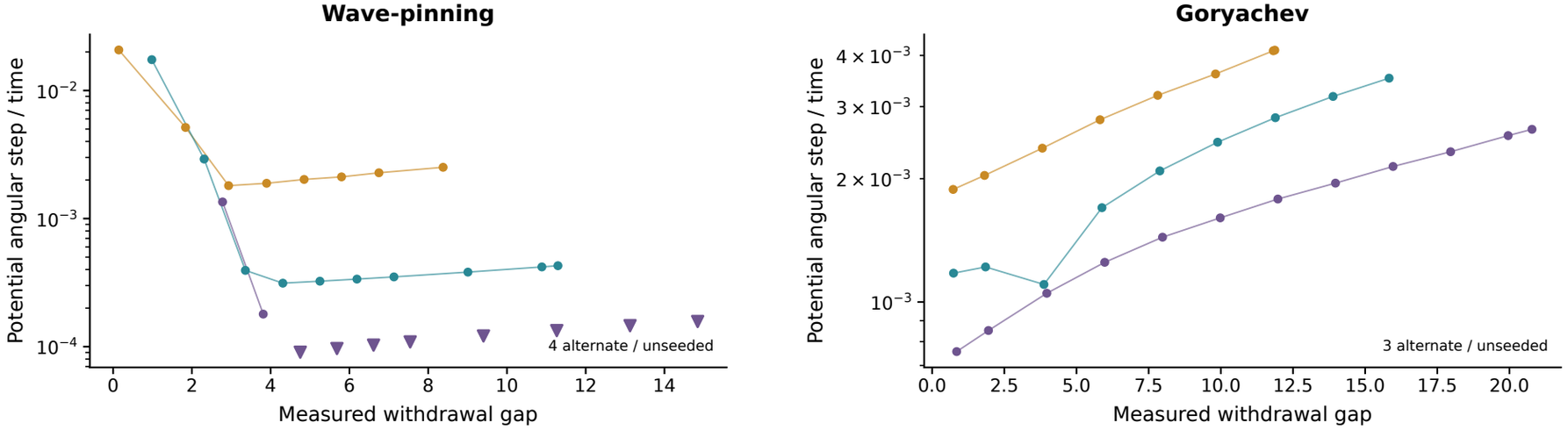
$$\theta = \frac{2\pi}{C}\left[g + \frac{w_{10} + s}{2}\right]$$

Measured widths enter the center step.
Handover and full-angle rotation are separate events.

E Fixed-angle cue placement: measured handover (solid), small-cell remap (dashed)

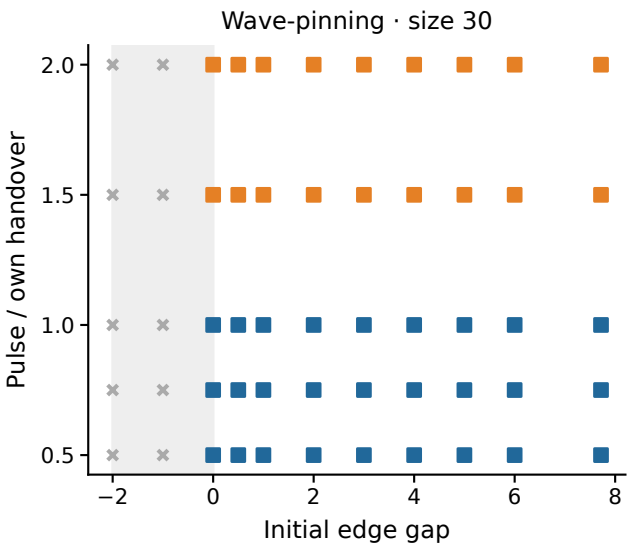


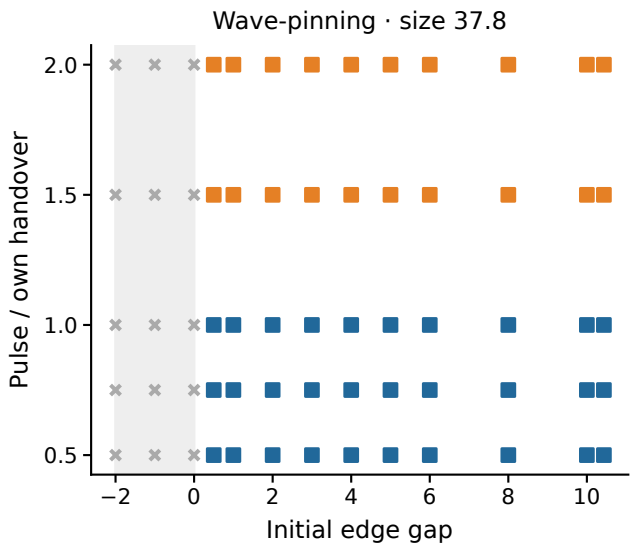
F Fixed physical scale: conditional geometric step / resolution time

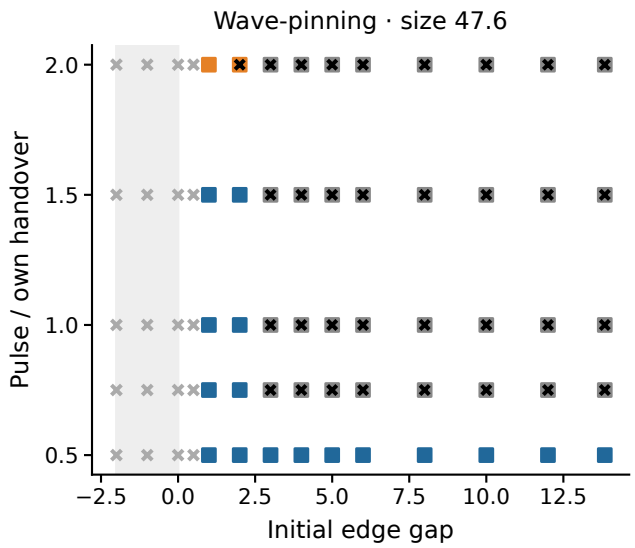


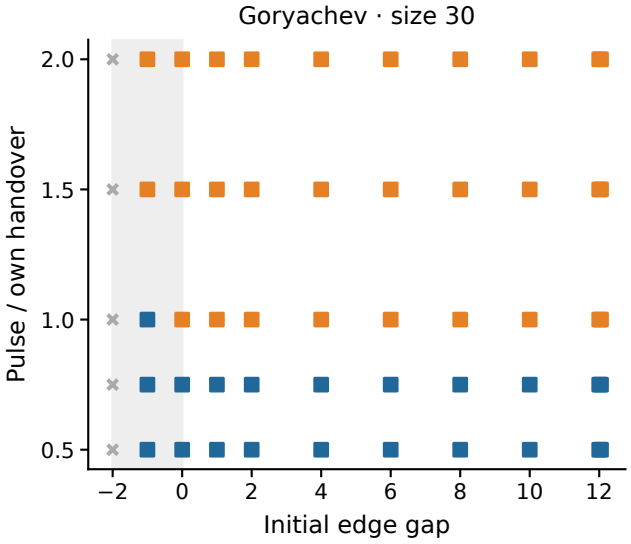
Dashed remaps stop at observed small-cell support. Insets measure actual global angle at handover, R≥0.05.
Potential step uses measured withdrawal centers and widths; direct heading-time tests and spatial outcomes are retained in the supplement.
These are native polarity-activity readouts; observed migration and conditional OU persistence estimates remain separate.

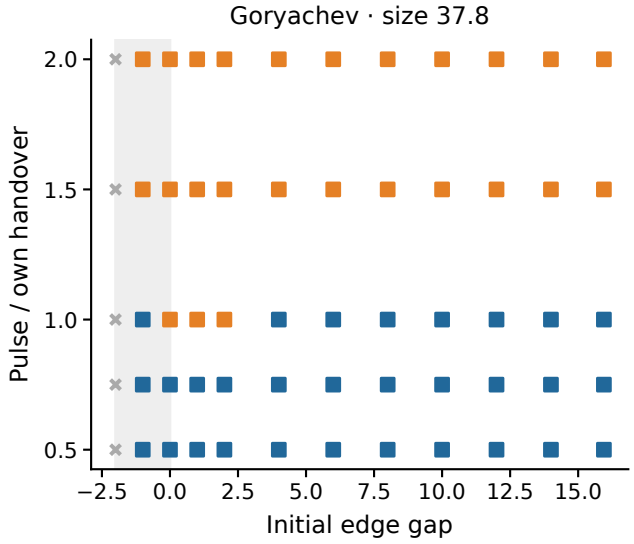
S1 · All 545 cue-withdrawal pulses and 34 unavailable own-handover placements

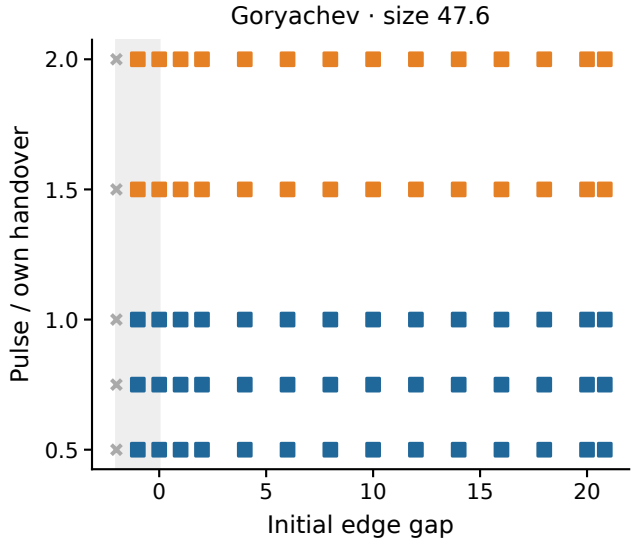


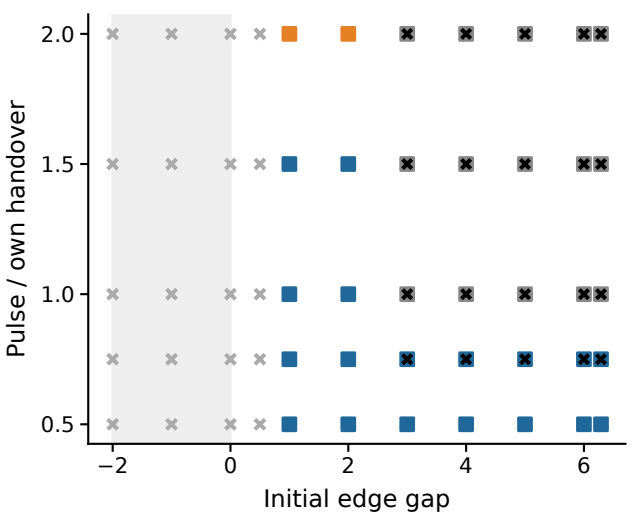


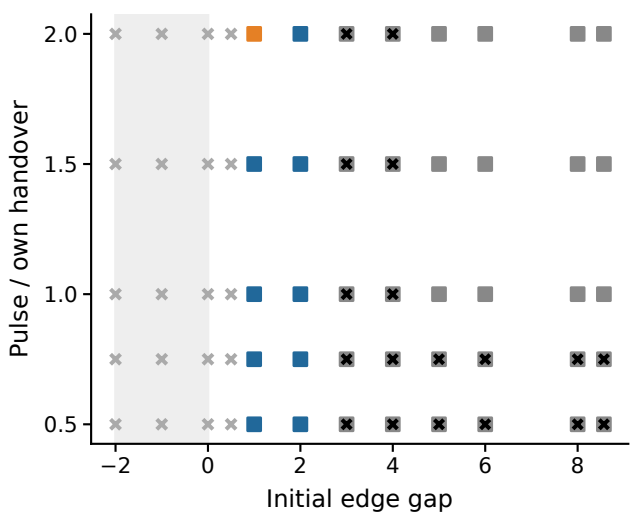


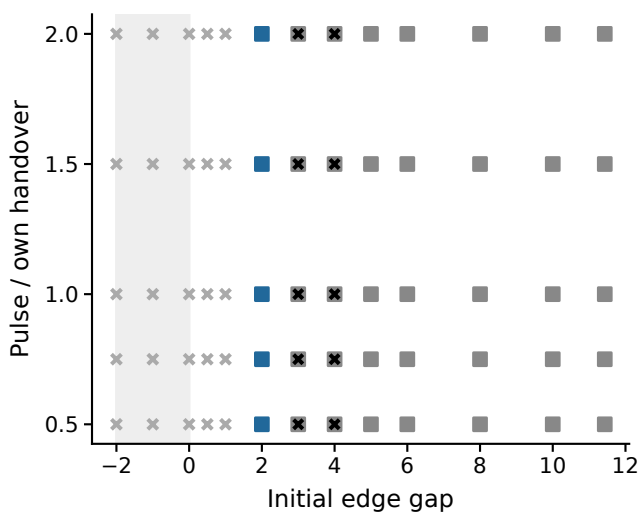


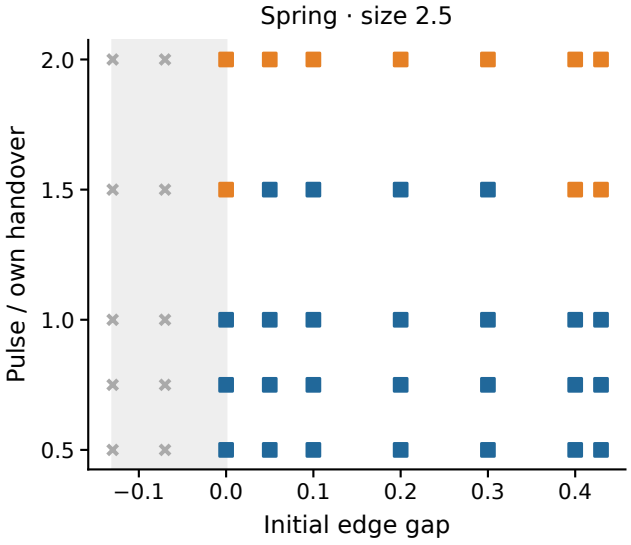


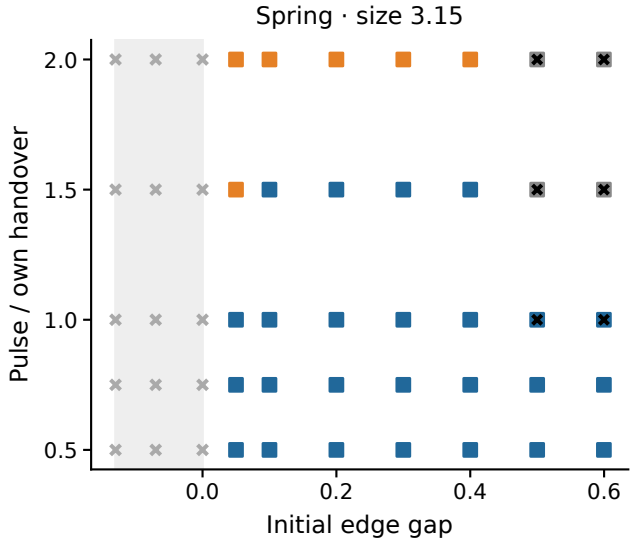


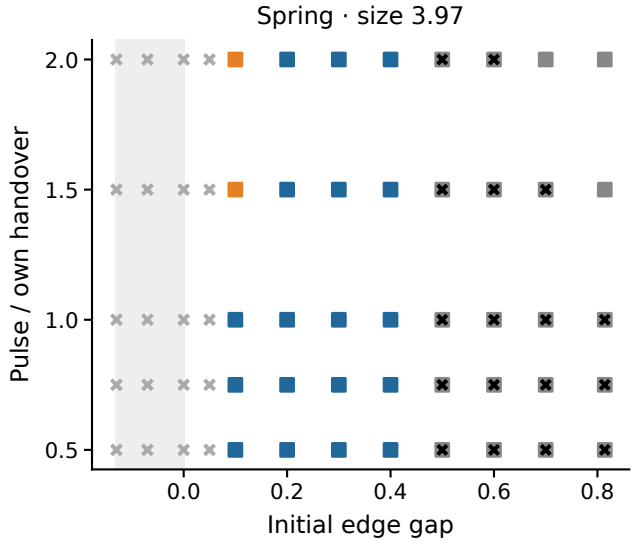








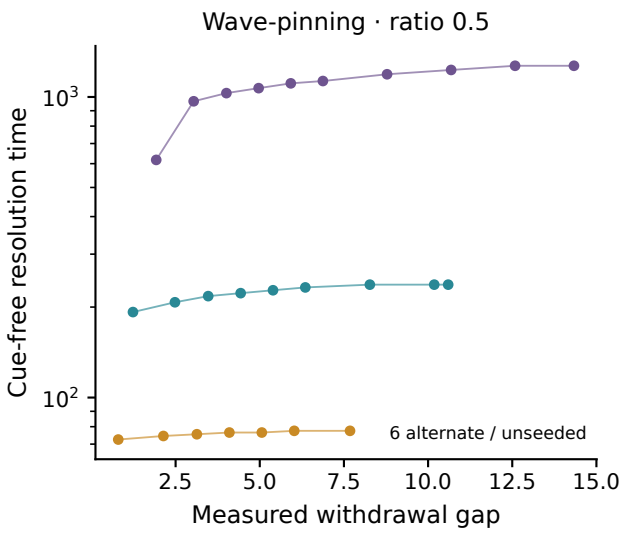


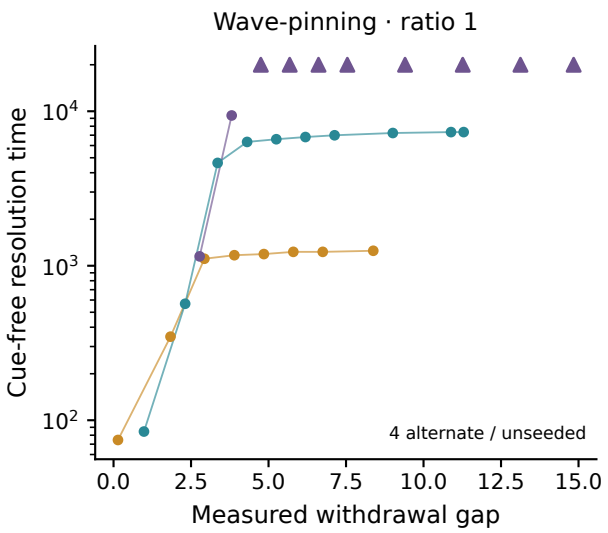


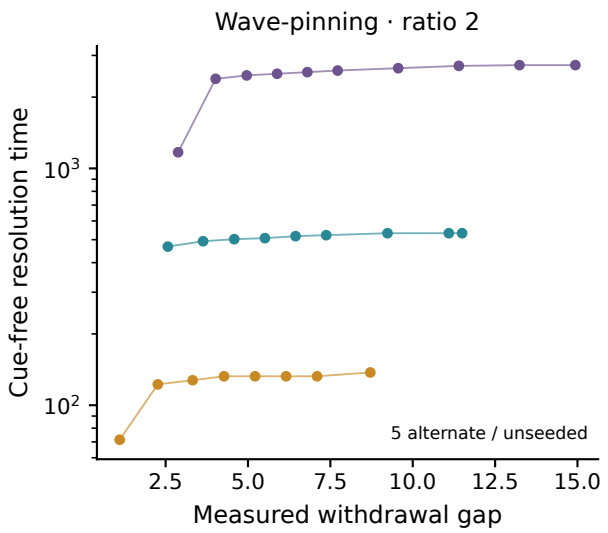
Blue: incumbent-associated final region; orange: challenger-associated; grey: unresolved. Grey ×: no true two-region handover.
Black × over a tested cell: finite native-field tail fails, independent of spatial fate. Shading: borderline placement.

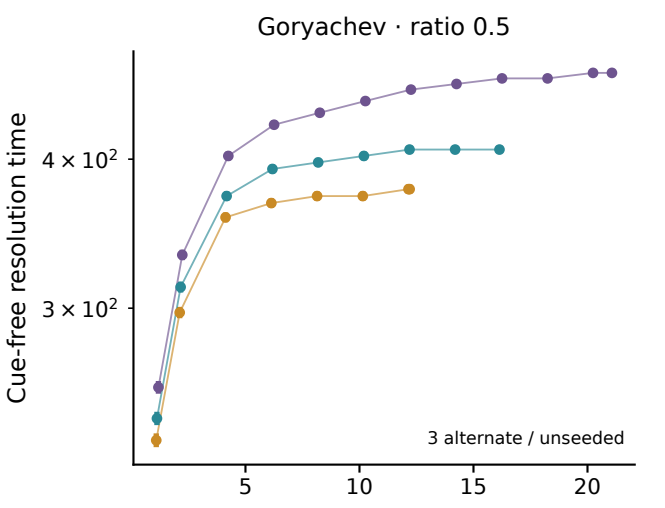
S2 · Cue-free resolution at all three complete-region withdrawal ratios

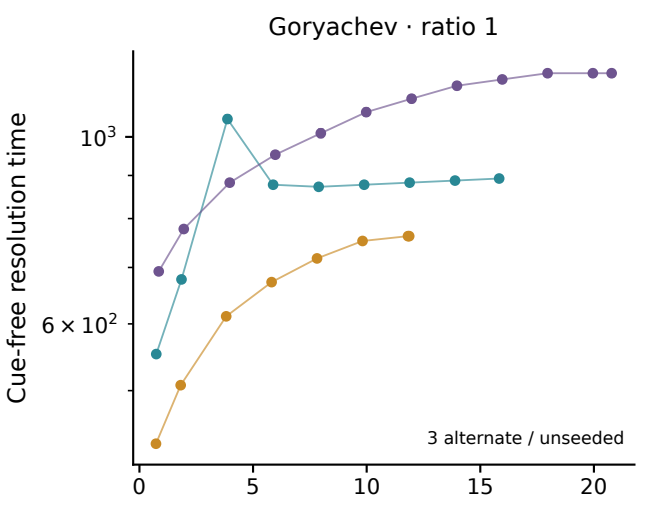
reference size · 1.26× native size label · 1.59× native size label

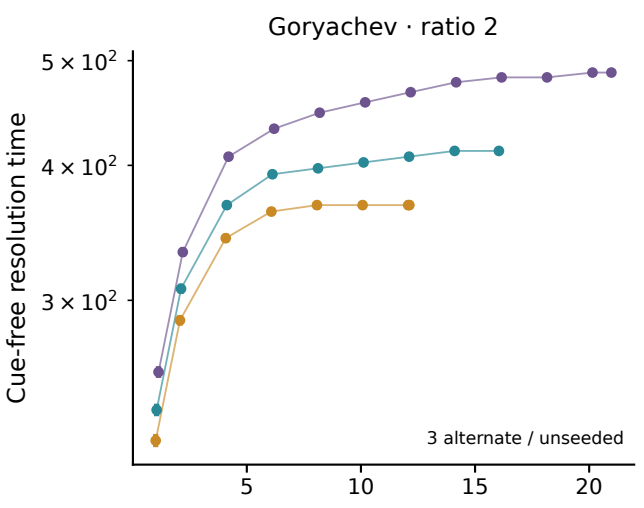


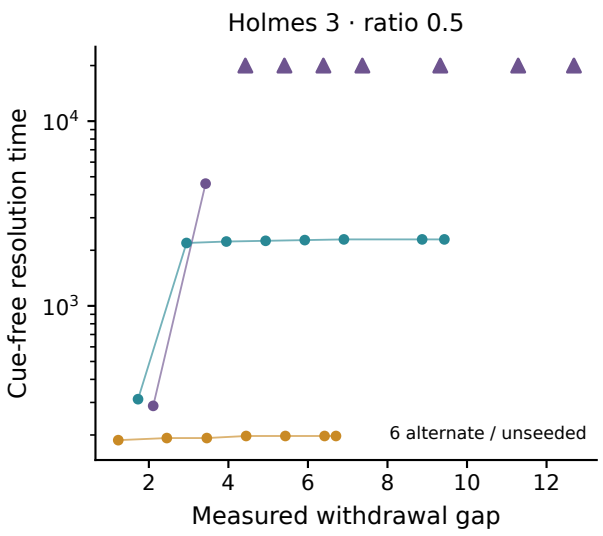


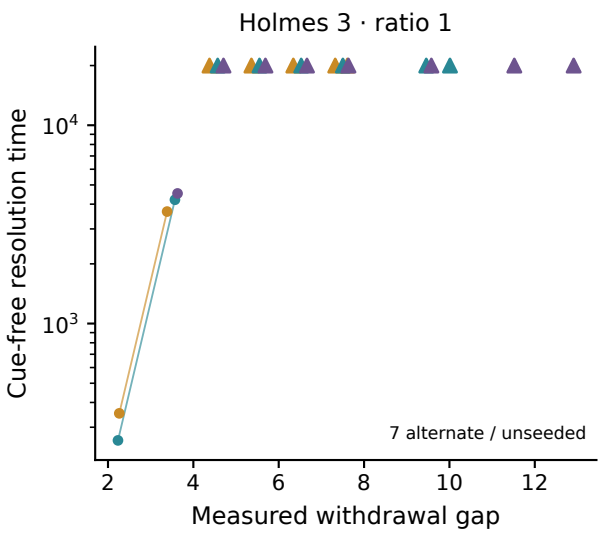


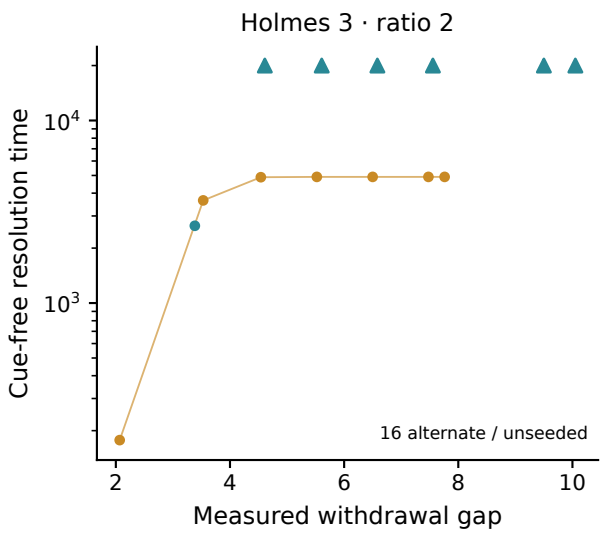


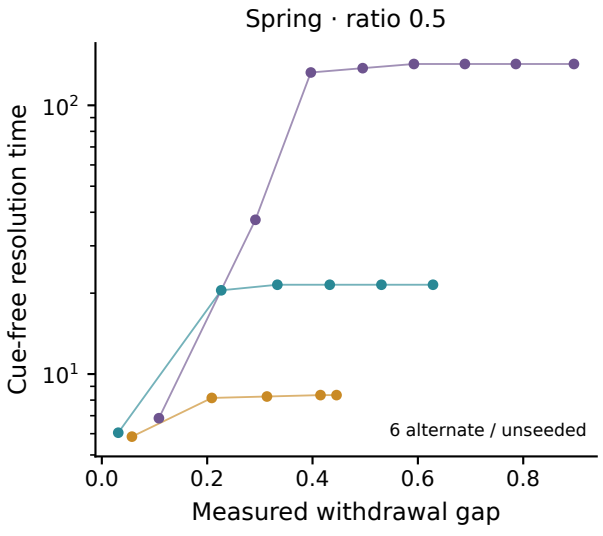


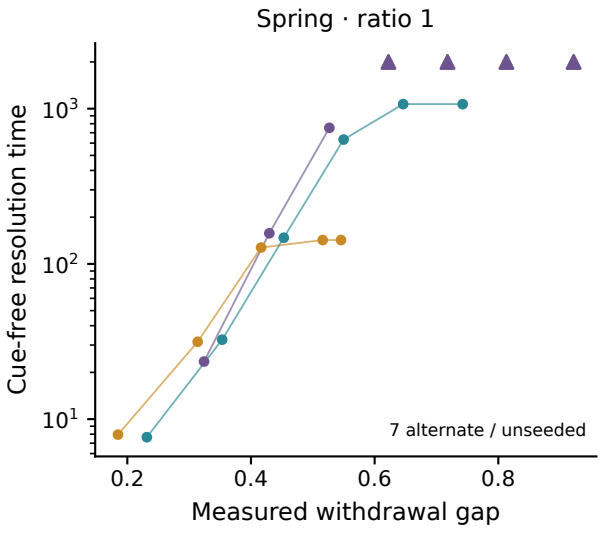


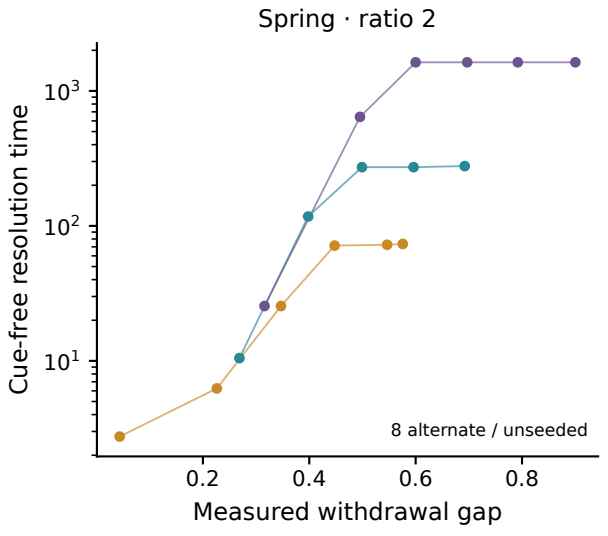




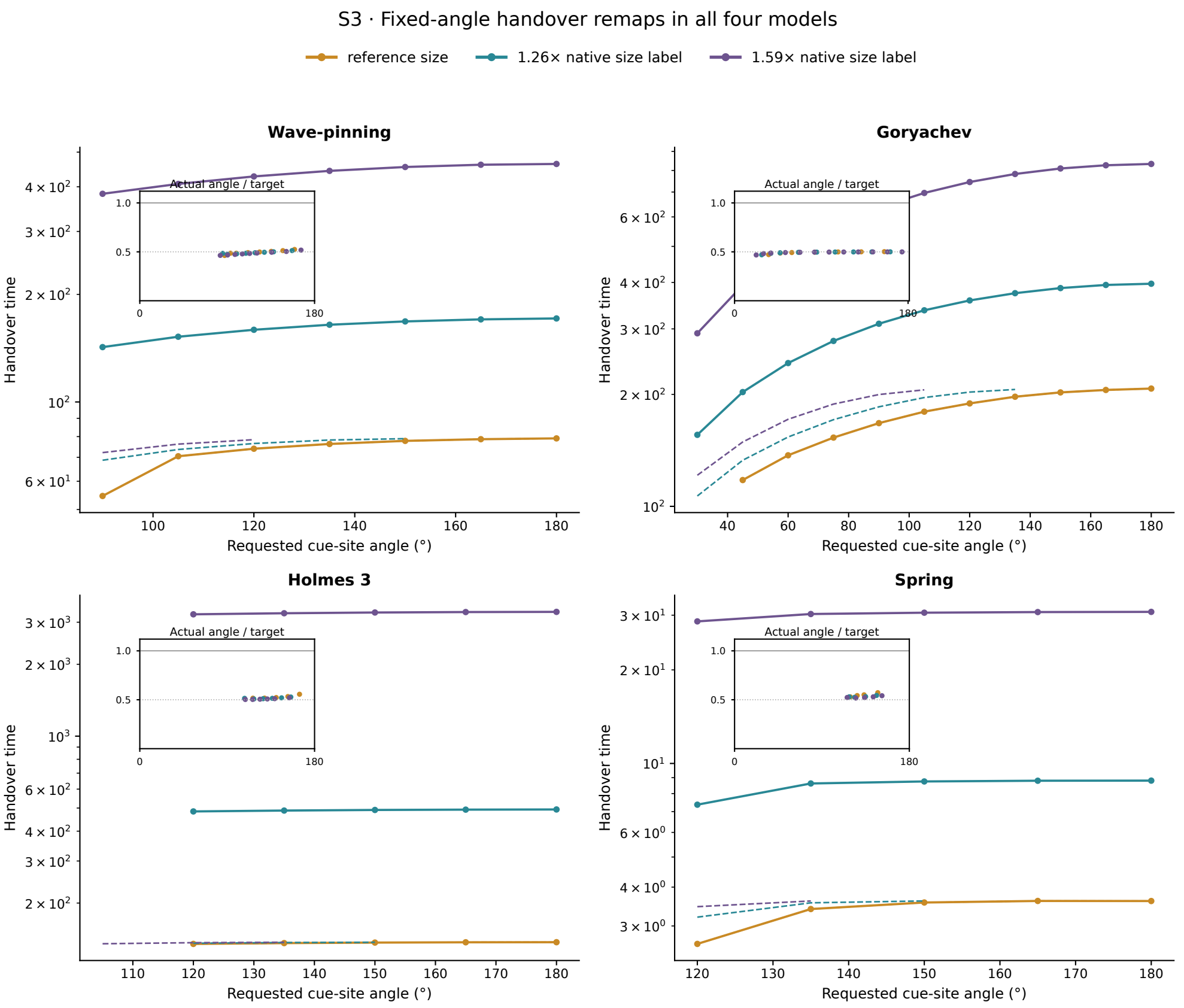








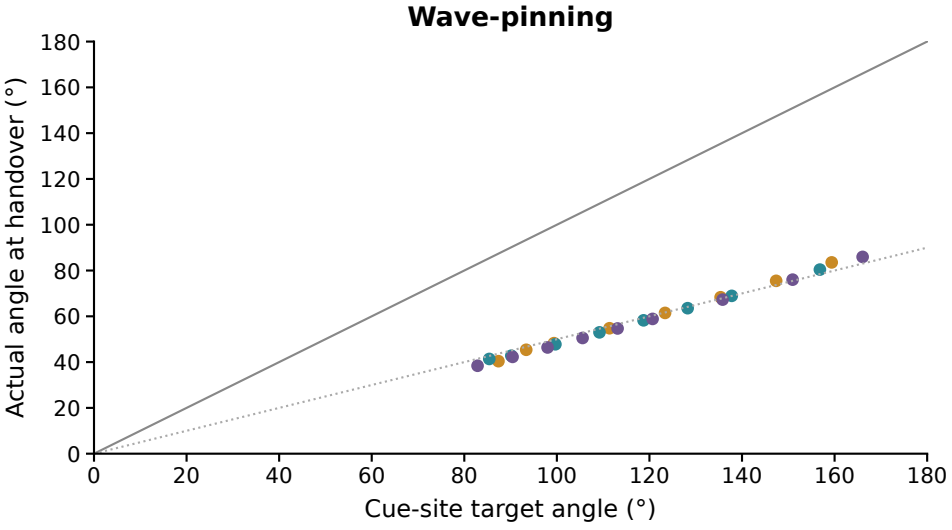
Primary gap: current global10%-contrast contours. ▲: resolution-time lower bound at completed horizon.
Alternate, merged and unseeded cases remain in the complete evidence table and native-history reader.

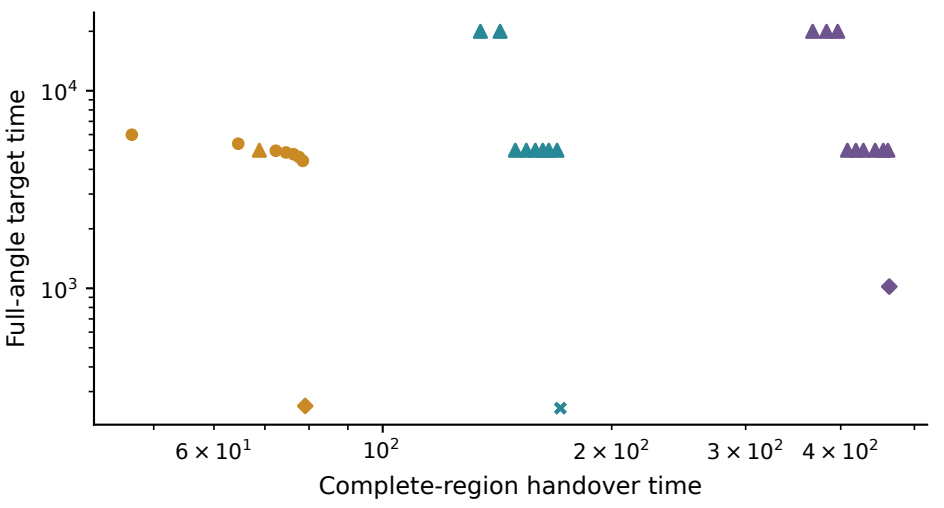

S3 · Fixed-angle handover remaps in all four models
reference size
1.26× native size label
1.59× native size label
Wave-pinning
Goryachev
Holmes 3
Spring
Handover time
Requested cue-site angle (°)
Actual angle / target
Solid: measured native size; dashed: reference-size handover curve evaluated at the larger cell's measured-width gap.
No extrapolation or interpolation through unavailable/censored neighbors. Insets: actual direction at handover / requested cue angle.

S4 · Handover typically reaches part of the requested angle; full-target timing is separate

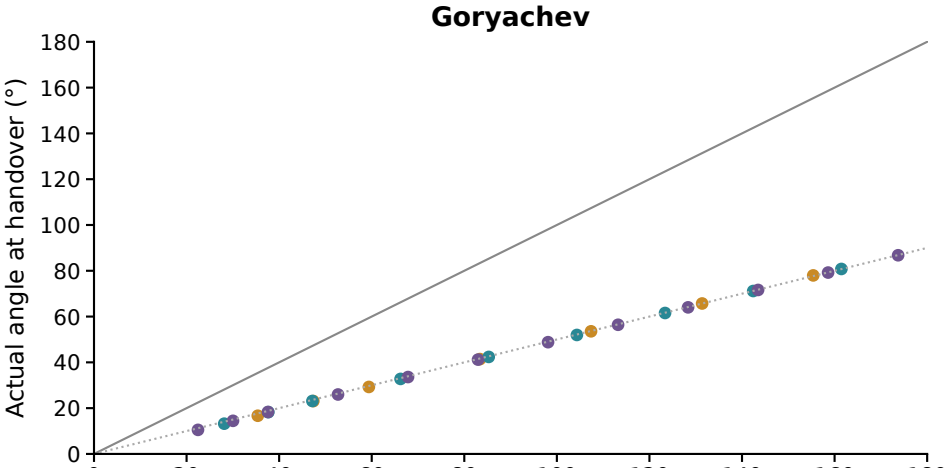
reference size · 1.26× native size label · 1.59× native size label

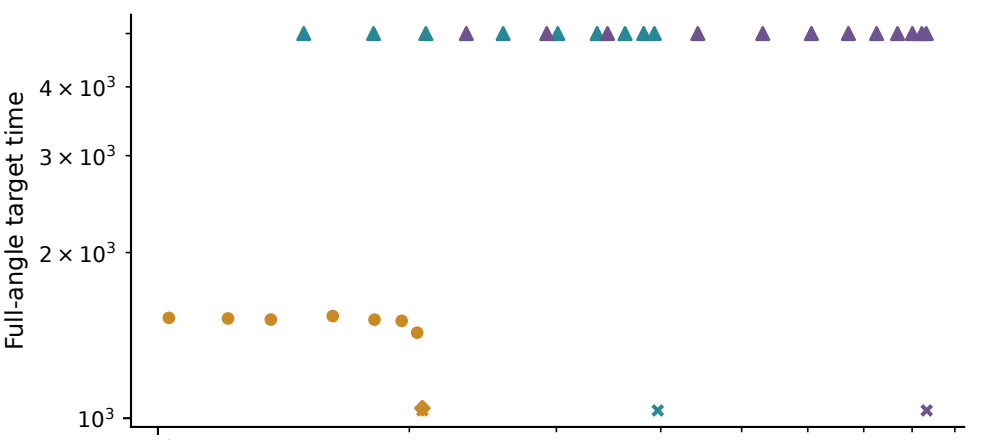
Wave-pinning



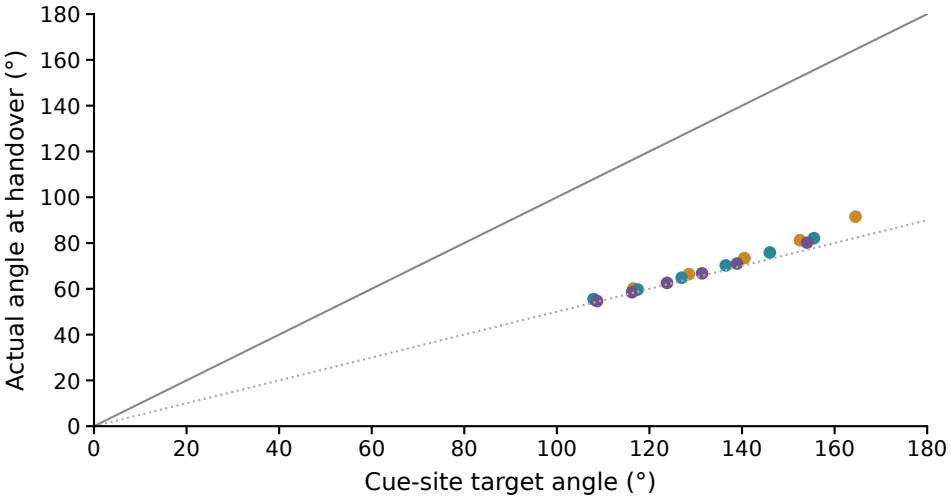


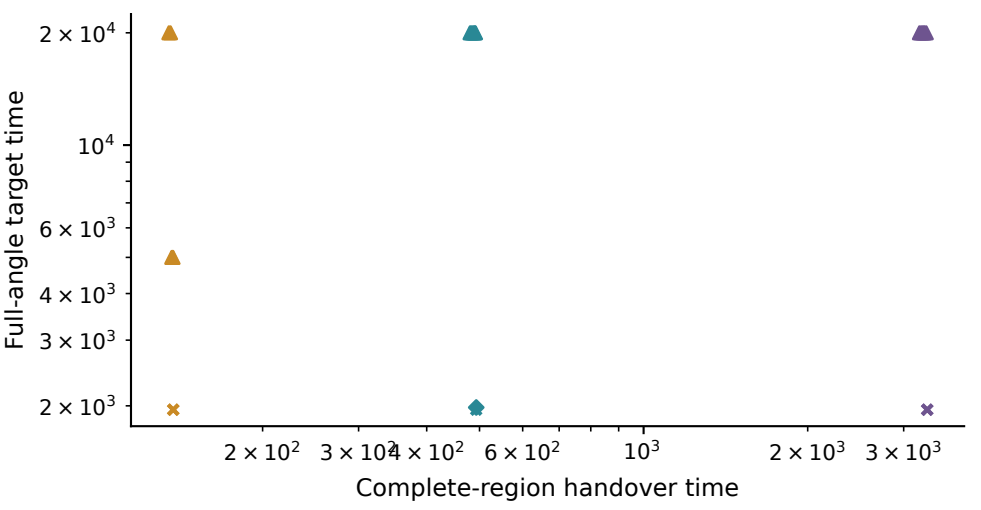
Goryachev



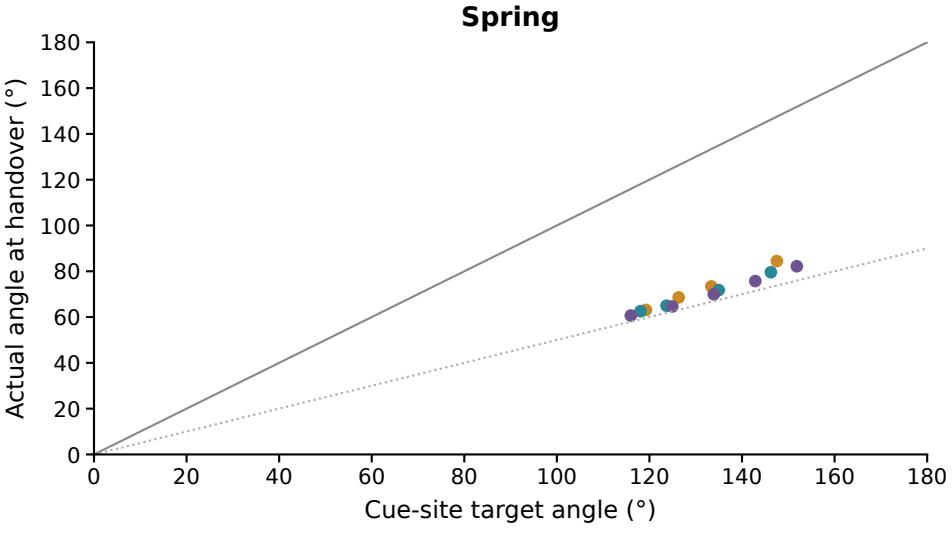


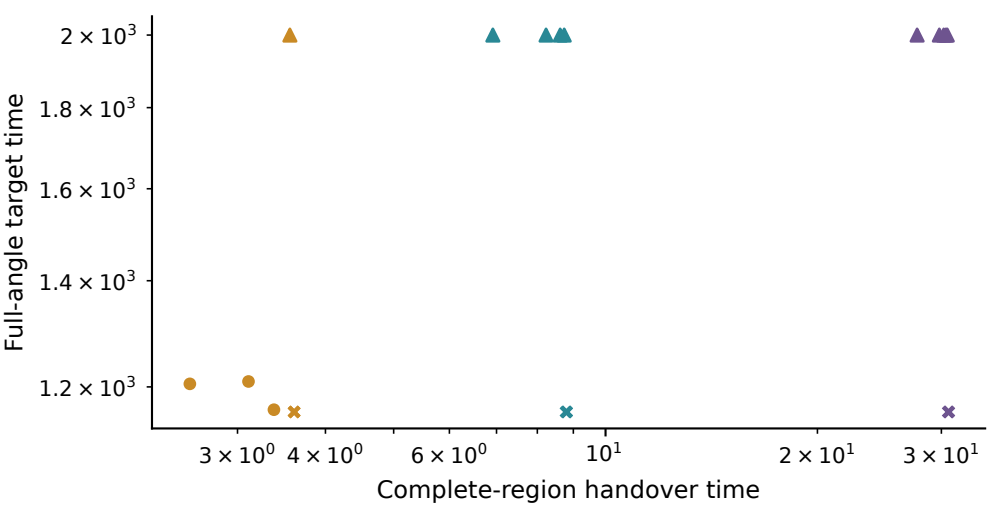
Holmes 3



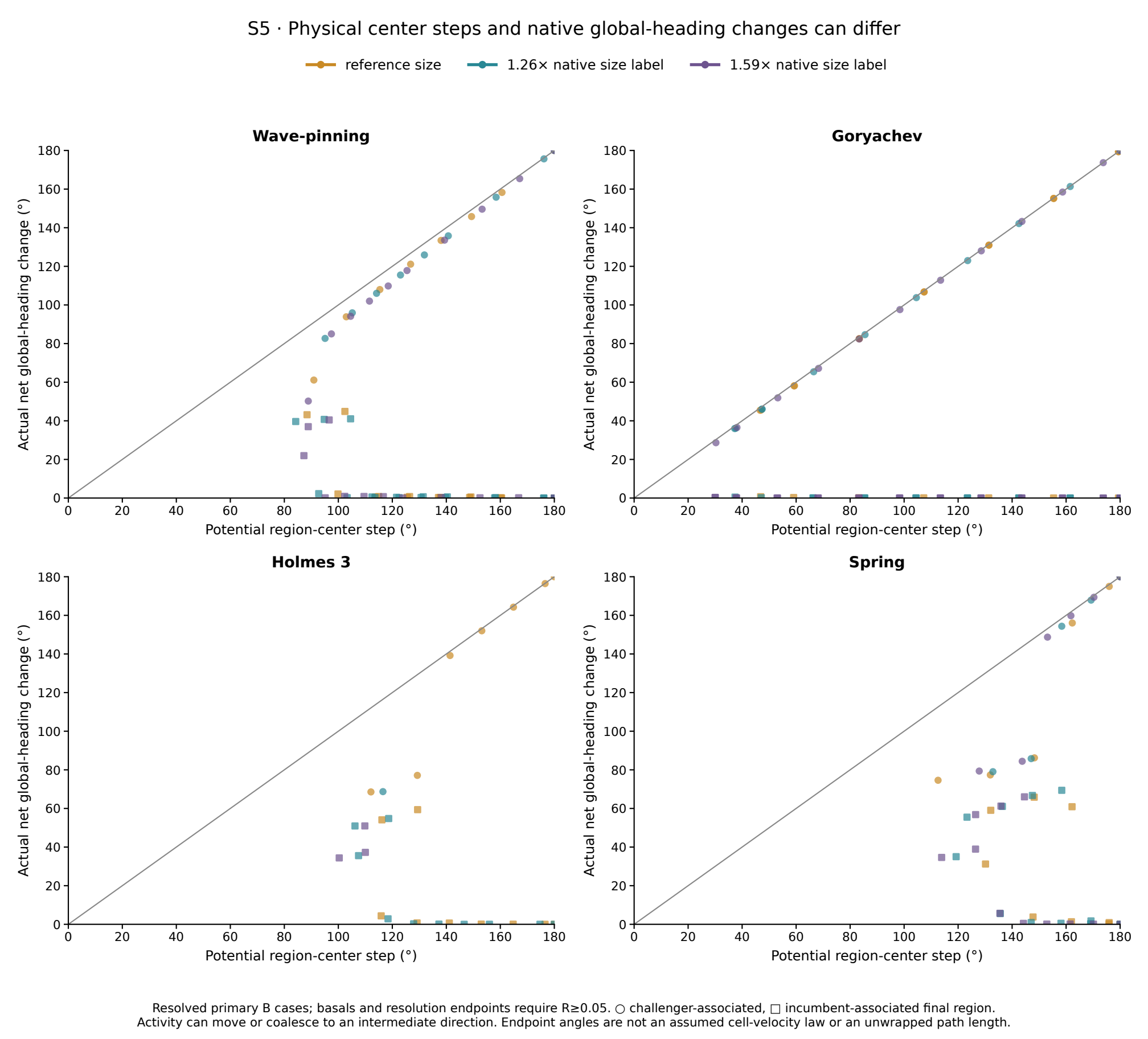


Spring

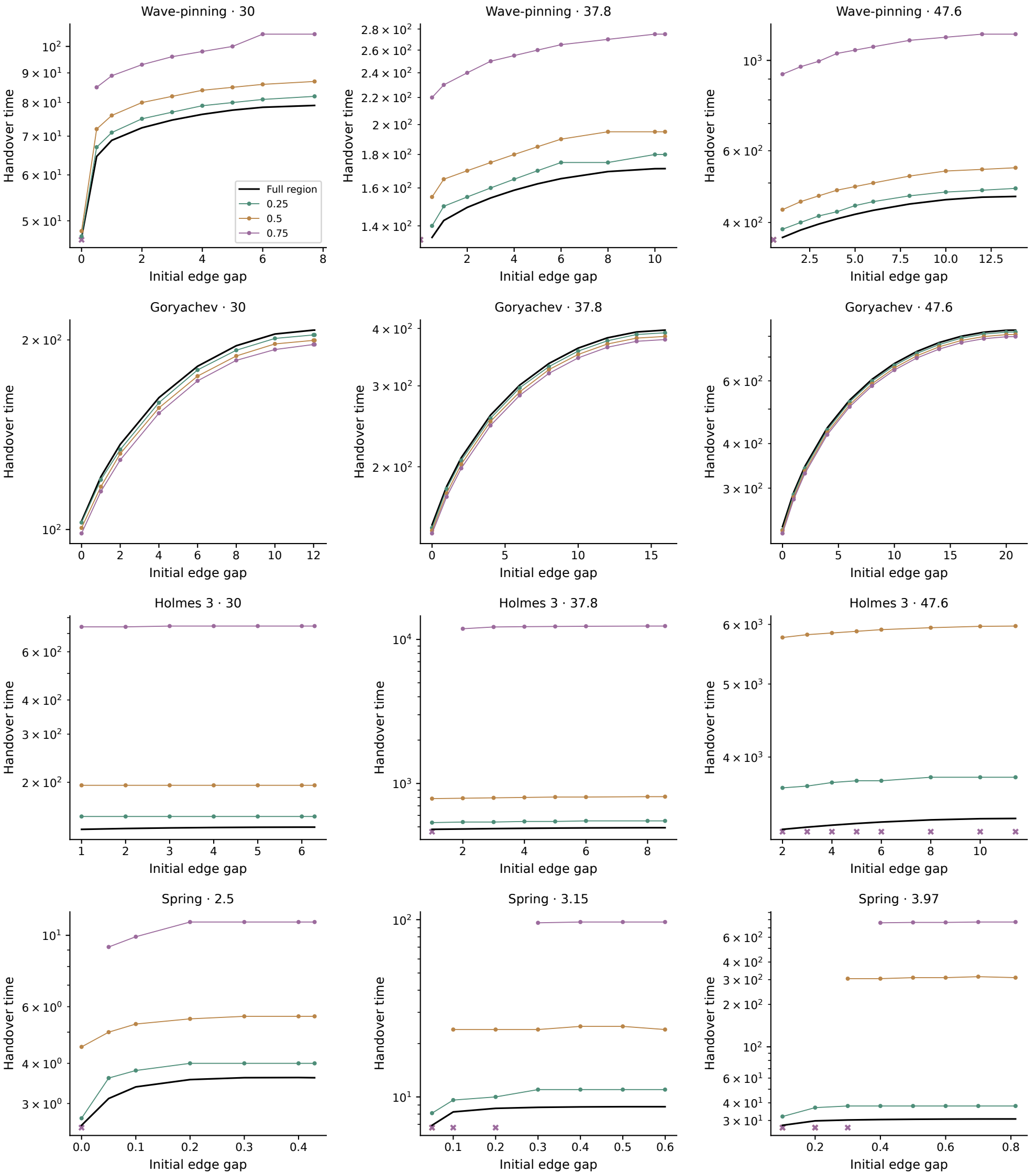




R=|M₁|/M₀≥0.05 at usable samples and intervening conservative complex-moment chords. ▲: finite-cap timing lower bound.
×: direction-quality/cancellation gap prevents a full timing bound. Exact-direction and0.1/0.2 gates plus50/90/99% targets are tabulated.

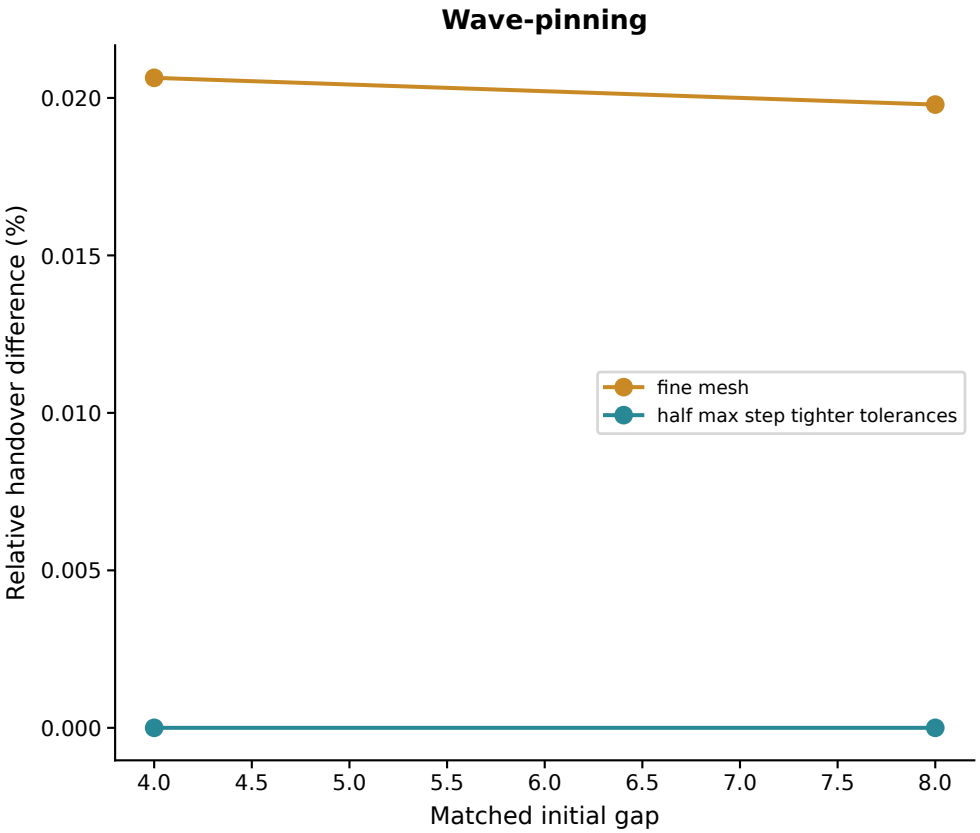

S5 · Physical center steps and native global-heading changes can differ
reference size
1.26× native size label
1.59× native size label
Wave-pinning
Goryachev
Holmes 3
Spring
Actual net global-heading change (°)
Potential region-center step (°)
0
20
40
60
80
100
120
140
160
180
Resolved primary B cases; basals and resolution endpoints require R≥0.05. ○ challenger-associated, □ incumbent-associated final region.
Activity can move or coalesce to an intermediate direction. Endpoint angles are not an assumed cell-velocity law or an unwrapped path length.

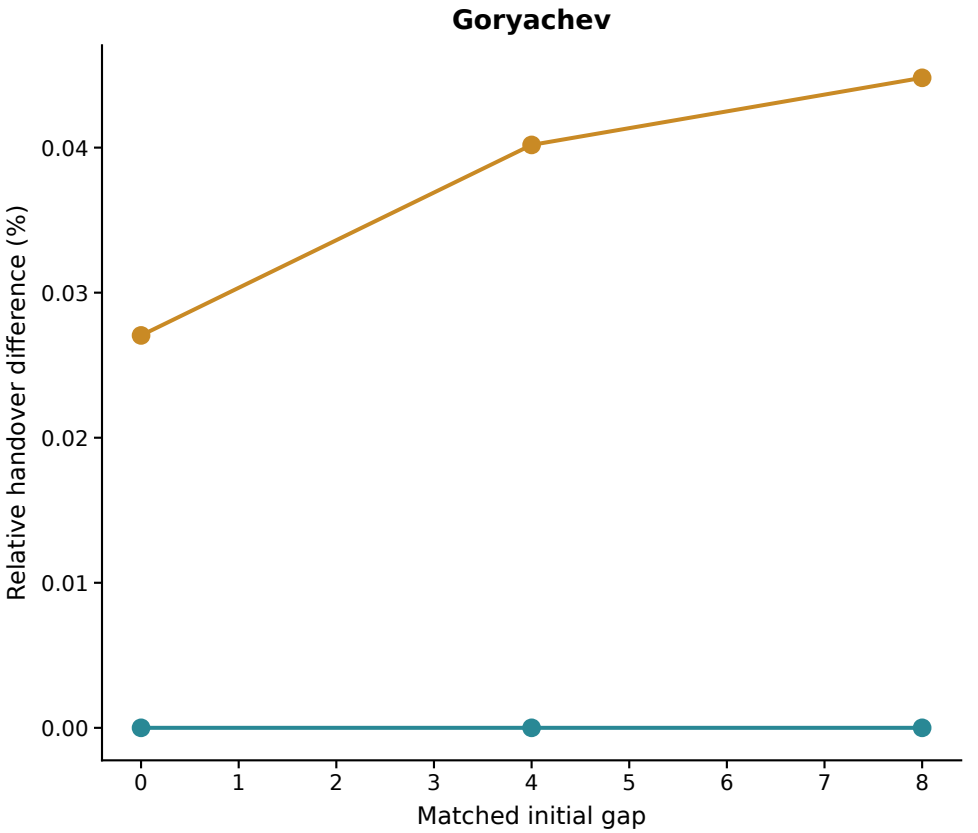
S6 · Amount sensitivity: local 25/50/75% peak contours versus complete-region excess



Original excess above each region's local-peak contour; watershed basins and common current floor are unchanged.
Times are retained native-sample upper brackets. ×: unavailable continuous sampled two-region crossing; no cap substitution.

S7 · Numerical, preparation and complete-source checks





Numerical evidence

Fresh native basals: 12 / 12 qualified
Long no-cue controls: 53,000 chemical / 5,300 Spring
Fine-mesh basals: 2 / 2 independently qualified
A1 refinement controls: 12 / 12 native audits
Observed control roots: 10 / 10 verified
A2 native audit: 545 / 545
B native audit: 381 / 381 (24 mirrors agree)
B mesh/step controls: 36 / 36 native audits

Readout and source gates

Native positivity and conserved-pool gates pass.
Scalar fate and field stationarity remain separate.
Continuous roots replay the original upper fields.
B controls retain all seed statuses and winners.
35 / 36 resolution brackets overlap.
Goryachev g=4, ratio 1, fine mesh: 1040–1060
versus 1080–1100 (about 4% timing sensitivity).
Full native atlas: 1,150 histories; all sources linked.